Logistic regression interpretation

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Logistic regression model

$Y_i =$ dengue status (0 = negative, 1 = positive)

$Age_i =$ age (in years)

Random component: $Y_i \sim Bernoulli(\pi_i)$

Systematic component: $\log\left(\frac{\pi_i}{1-\pi_i}\right) = \beta_0 + \beta_1\, Age_i$

Fitting the logistic regression model

$$Y_i \sim Bernoulli(\pi_i)$$

$$\log\left(\frac{\pi_i}{1-\pi_i}\right) = \beta_0 + \beta_1\, Age_i$$

```
m1 <- glm(Dengue ~ Age, data = dengue,
          family = binomial)
summary(m1)
```

Fitting the logistic regression model

$$Y_i \sim Bernoulli(\pi_i)$$

$$\log\left(\frac{\pi_i}{1-\pi_i}\right) = \beta_0 + \beta_1\ Age_i$$

```
m1 <- glm(Dengue ~ Age, data = dengue,
          family = binomial)
summary(m1)
```

```
...
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept) -2.454345   0.075068  -32.70   <2e-16 ***
## Age          0.217312   0.008826   24.62   <2e-16 ***
...
```

Last time: Activity

$$\log\left(\frac{\widehat{\pi}_i}{1-\widehat{\pi}_i}\right) = -2.45 + 0.22\ Age_i$$

> Does the probability of dengue increase or decrease with age?

Activity

$$\log\left(\frac{\widehat{\pi}_i}{1-\widehat{\pi}_i}\right) = -2.45 + 0.22\ Age_i$$

> What are the predicted odds of dengue for a 5 year old patient?

Activity

$$\log\left(\frac{\widehat{\pi}_i}{1-\widehat{\pi}_i}\right) = -2.45 + 0.22\ Age_i$$

For a 5 year old patient: is the predicted *probability* of dengue > 0.5, < 0.5, or $= 0.5$?

Calculating probability

$$\log\left(\frac{\widehat{\pi}_i}{1-\widehat{\pi}_i}\right) = -2.45 + 0.22\ Age_i$$

Predicted *probability* of dengue for a 5 year old patient?

Activity

Work on the activity (handout) with a neighbor, then we will discuss as a class.

Activity

$$\log\left(\frac{\widehat{\pi}_i}{1-\widehat{\pi}_i}\right) = -2.45 + 0.22\ Age_i$$

What is the predicted probability of dengue for a 10 year old patient?

Activity

$$\log\left(\frac{\widehat{\pi}_i}{1-\widehat{\pi}_i}\right) = -2.45 + 0.22\ Age_i$$

Suppose we want to identify patients for whom the predicted probability of dengue is at least 0.5. What age range should we focus on?

Activity

$$\log\left(\frac{\widehat{\pi}_i}{1-\widehat{\pi}_i}\right) = -2.45 + 0.22\ Age_i$$

> Compare the odds of dengue for a 12 year old patient to the odds of dengue for an 11 year old patient. What do you notice?

Interpretation

$$\log\left(\frac{\widehat{\pi}_i}{1-\widehat{\pi}_i}\right) = -2.45 + 0.22\ Age_i$$

Slope:

- A one-year increase in age is associated with an increase of 25% in the *odds* of dengue.

Intercept:

Interpretation

$$\log\left(\frac{\widehat{\pi}_i}{1-\widehat{\pi}_i}\right) = -2.45 + 0.22\ Age_i$$

Slope:

- A one-year increase in age is associated with an increase of 25% in the *odds* of dengue.

Intercept:

- when age is 0, the estimated odds of dengue is $e^{-2.45} = 0.086$

Adding more variables

Now let's add WBC as a variable to the model:

```
m2 <- glm(Dengue ~ Age + WBC, data = dengue,
          family = binomial)
summary(m2)
```

$$\log\left(\frac{\widehat{\pi}_i}{1-\widehat{\pi}_i}\right) = 0.34 + 0.15\ Age_i - 0.31 WBC_i$$

> How should I interpret each coefficient in the fitted model?

Adding more variables

What about a *categorical* predictor?

```
m3 <- glm(Dengue ~ Age + Abdominal, data = dengue,
          family = binomial)
summary(m3)
```

$$\log\left(\frac{\widehat{\pi}_i}{1-\widehat{\pi}_i}\right) = -0.25 + 0.22\ Age_i + 0.21 Abdominal_i$$

> How should I interpret the coefficient on abdominal pain?

Interpretation

$$\log\left(\frac{\widehat{\pi}_i}{1-\widehat{\pi}_i}\right) = -2.45 + 0.22\ Age_i$$

Slope:

- A one-year increase in age is associated with an increase of 25% in the *odds* of dengue.

> Can we interpet the slope in terms of the the *probability* (rather than the odds or log odds)?

For next week

Reading: in the textbook, read

- sections 6.7.1 - 6.7.3
- sections 2.2 - 2.4